COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure Algorithms	**SECTION:**	CPE21S2
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SEATWORK 10.1 Graph

1. PROCEDURE OUTPUT

1. Define the following: (10)

A. Graph

- A mathematical structure consisting of a set of objects (vertices or nodes) where some pairs of objects are connected by links (edges).

B. Vertex

- A fundamental unit or point that forms a graph, representing an individual entity or data point.

C. Nodes

- An alternative term for vertices, commonly used in computer science and network contexts to signify a point of connection.

D. Weight

- A numerical value assigned to an edge in a graph, often representing cost, distance, capacity, or resistance between two connected vertices.

E. Path

- A sequence of edges that connects a sequence of distinct vertices, allowing traversal from a starting vertex to a destination vertex.

F. Directed graph

- A graph where the edges have a specific orientation or direction, meaning they point from one vertex to another (often represented by arrows).

G. Connected graph

- A graph in which there is a path between every pair of vertices, meaning no vertex or

group of vertices is completely isolated.

H. Directed cyclic

- A directed graph that contains at least one cycle—a path that starts and ends at the same vertex by following the directed edges.

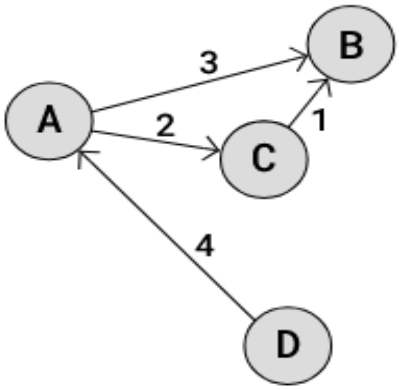
I. Adjacency matrix

- A two-dimensional square matrix used to represent a graph, where the rows and columns represent vertices, and the cell values indicate whether an edge exists between them (and its weight, if applicable).

J. Adjacency List

- A collection of unordered lists used to represent a graph, where each vertex has a dedicated list that stores all the other vertices it is directly connected to.

2. Identify the parts of the following graph. (10)



- Nodes: {A, B, C, D}

Edges	Weight
A to B	3
A to C	2
C to B	1
D to A	4

3. Plot the adjacency matrix of the graph above. (10 pts)

	A	**B**	**C**	**D**
A	0	3	2	0
B	0	0	0	0
C	0	1	0	0
D	4	0	0	0

4. Create a program in C++ program to create a graph that looks like in the above figure.

```
Code:
#include <iostream>
#include <vector>
#include <iomanip>

using namespace std;

int main() {
   // Number of nodes (A=0, B=1, C=2, D=3)
   const int NUM_NODES = 4;
   char nodeLabels[] = {'A', 'B', 'C', 'D'};

   // Adjacency matrix based on item 3
   // Rows/Columns: A, B, C, D
   int adjMatrix[NUM_NODES][NUM_NODES] = {
      {0, 3, 2, 0}, // Edges from A
      {0, 0, 0, 0}, // Edges from B
      {0, 1, 0, 0}, // Edges from C
      {4, 0, 0, 0}  // Edges from D
   };

   // Print Nodes
   cout << "Nodes: {A, B, C, D}\n\n";

   // Print Directed Edges and Weights (from item 2)
   cout << "Edges and Weights:\n";
   cout << "--------------------\n";
   for (int i = 0; i < NUM_NODES; i++) {
      for (int j = 0; j < NUM_NODES; j++) {
         if (adjMatrix[i][j] != 0) {
            cout << nodeLabels[i] << " to " << nodeLabels[j]
                 << " -> Weight: " << adjMatrix[i][j] << "\n";
         }
      }
   }

   cout << "\n";

   // Print Adjacency Matrix (from item 3)
   cout << "Adjacency Matrix:\n";
   cout << "    A  B  C  D\n";
   cout << "   -------------\n";
   for (int i = 0; i < NUM_NODES; i++) {
```

```
        cout << nodeLabels[i] << " |";
        for (int j = 0; j < NUM_NODES; j++) {
            cout << setw(3) << adjMatrix[i][j];
        }
        cout << "\n";
    }

    return 0;
}
```

5. Differentiate a BFS vs DFS types of search.

Breadth-First Search (BFS)

- **Advantages of BFS**
    - **Optimal Solution -** BFS is guaranteed to find the shortest path between two nodes in an unweighted graph.
    - **Completeness -** BFS is complete, meaning it will find the solution if it exists.
    - **Uniform cost search -** BFS can be modified to search for the shortest path in a weighted graph by replacing the queue data structure with a priority queue.
    - **Level-wise traversal -** BFS visits all the nodes at a given level before moving to the next level. This approach makes it useful in certain scenarios such as finding the shortest path or exploring a game state.

- **Disadvantages of BFS**
    - **Space complexity -** BFS can be memory-intensive, especially for large graphs or trees. It needs to store all the visited nodes in a queue data structure, which can take up a lot of memory.
    - **Time complexity -** BFS can be slow for graphs with a large number of nodes or edges.

Depth-First Search (DFS)

- **Advantages of DFS**
    - **Memory-efficient -** DFS uses less memory than BFS. Only the path from the root node to the current node is needed.
    - **Time Efficient -** For graphs with many nodes or edges, DFS may be faster than BFS. Its time complexity is O(b^m), where b is the branching factor and m is the maximum graph depth.
    - **Depth-first search -** DFS explores the deepest path first. This approach makes it useful

in certain scenarios, such as finding the maximum depth of a tree or graph.

- **Disadvantages of DFS**
    - **Completeness -** DFS is not guaranteed to find the solution if it exists. If the graph has cycles, it may loop forever.
    - **Non-optimal solution -** DFS may not find the shortest path between two nodes in an unweighted graph. It may find a longer path before finding the shortest path.
    - **Local minimum -** DFS may get stuck in a local minimum in a weighted graph.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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